

1 de Febrero del 2026

PRACTICA 1

Bases de datos distribuidas
Gerardo Jiménez Haro

Jiovanni Lopez

- Primero descargamos e instalamos actualizaciones generales del sistema

```
3root@vbox:/home/dapa#
root@vbox:/home/dapa# sudo apt update && apt upgrade -y
Hit:1 http://deb.debian.org/debian bookworm InRelease
Hit:2 http://security.debian.org/debian-security bookworm-security InRelease
Hit:3 http://deb.debian.org/debian bookworm-updates InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

- Haciendo uso de wget, descargamos el archivo necesario para la instalación de mysql server

```
root@vbox:/tmp# wget https://dev.mysql.com/get/mysql-apt-config_0.8.36-1_all.deb
--2026-02-01 12:37:54-- https://dev.mysql.com/get/mysql-apt-config_0.8.36-1_all.deb
Resolving dev.mysql.com (dev.mysql.com)... 184.86.165.250, 2600:141c:2800:a83::2e31, 2600:141c:2800:a81::2e31
Connecting to dev.mysql.com (dev.mysql.com)|184.86.165.250|:443... connected.
HTTP request sent, awaiting response... 302 Moved Temporarily
Location: https://repo.mysql.com//mysql-apt-config_0.8.36-1_all.deb [following]
--2026-02-01 12:37:54-- https://repo.mysql.com//mysql-apt-config_0.8.36-1_all.deb
Resolving repo.mysql.com (repo.mysql.com)... 23.4.56.100, 2600:1404:200:180::1d68, 2600:1404:200:18b::1d68
Connecting to repo.mysql.com (repo.mysql.com)|23.4.56.100|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 18128 (18K) [application/x-debian-package]
Saving to: 'mysql-apt-config_0.8.36-1_all.deb'

mysql-apt-config_0. 100%[=====>] 17.70K --.-KB/s in 0.001s

2026-02-01 12:37:55 (28.1 MB/s) - 'mysql-apt-config_0.8.36-1_all.deb' saved [18128/18128]
```

- Una vez descargado lo instalamos haciendo uso de la instrucción dpkg

```
root@vbox:/tmp# sudo dpkg -i mysql-apt-config_0.8.36-1_all.deb
Selecting previously unselected package mysql-apt-config.
(Reading database ... 151273 files and directories currently installed.)
Preparing to unpack mysql-apt-config_0.8.36-1_all.deb ...
Unpacking mysql-apt-config (0.8.36-1) ...
Setting up mysql-apt-config (0.8.36-1) ...
```

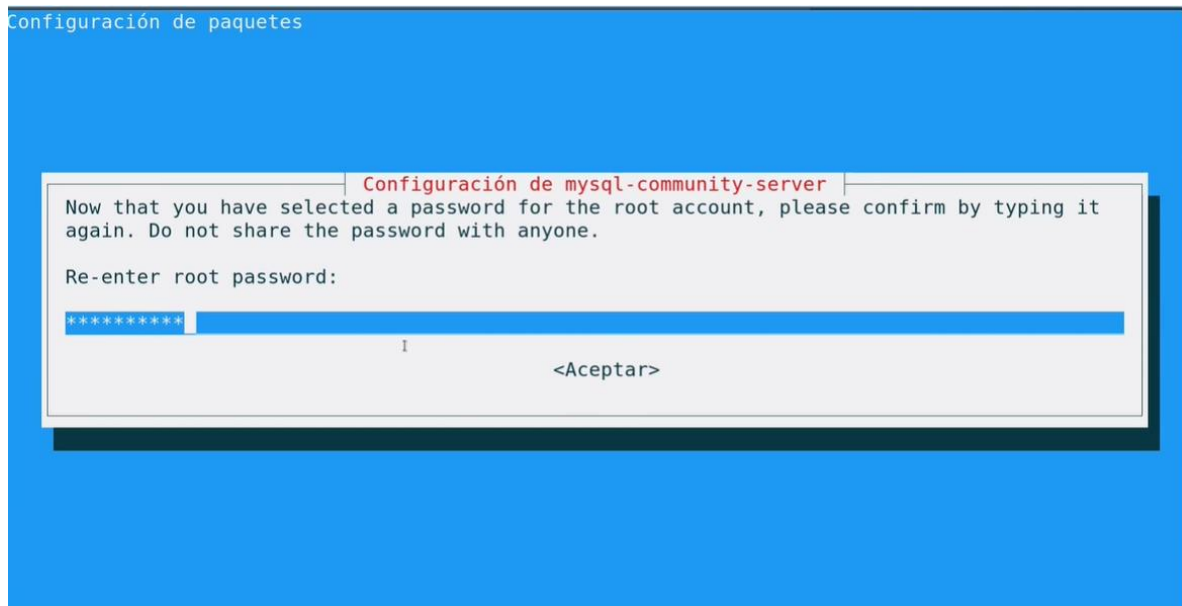
- Solo esperamos a que termine con la instalación

```

root@vbox:/tmp# sudo apt install mysql-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libaio1 libmecab2 mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client
  mysql-common mysql-community-client mysql-community-client-core
  mysql-community-client-plugins mysql-community-server
  mysql-community-server-core
The following NEW packages will be installed:
  libaio1 libmecab2 mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client
  mysql-common mysql-community-client mysql-community-client-core
  mysql-community-client-plugins mysql-community-server
  mysql-community-server-core mysql-server
0 upgraded, 13 newly installed, 0 to remove and 0 not upgraded.
Need to get 46.9 MB of archives.
After this operation, 355 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://deb.debian.org/debian bookworm/main amd64 libaio1 amd64 0.3.113-4 [
13.4 kB]
Get:2 http://deb.debian.org/debian bookworm/main amd64 libmecab2 amd64 0.996-14+
b14 [222 kB]
Get:3 http://deb.debian.org/debian bookworm/main amd64 mecab-utils amd64 0.996-1

```

- Una vez termine el proceso nos aparecerá esta pantalla que nos pedirá una contraseña segura para la cuenta root



- Comprobamos que mysql este funcionando correctamente haciendo uso del comando `systemctl status`

```
root@vbox:/tmp# sudo systemctl status
• mysql.service - MySQL Community Server
  Loaded: loaded (/lib/systemd/system/mysql.service; enabled; preset: enable>
  Active: active (running) since Sun 2026-02-01 12:42:03 CST; 2min 33s ago
    Docs: man:mysqld(8)
          http://dev.mysql.com/doc/refman/en/using-systemd.html
 Main PID: 4669 (mysqld)
   Status: "Server is operational"
    Tasks: 34 (limit: 2277)
  Memory: 426.6M
     CPU: 2.411s
  CGroup: /system.slice/mysql.service
          └─4669 /usr/sbin/mysqld

Feb 01 12:41:56 vbox systemd[1]: Starting mysql.service - MySQL Community Serve>
Feb 01 12:42:03 vbox systemd[1]: Started mysql.service - MySQL Community Server.
lines 1-15/15 (END)
```

- Ahora, similar que con mysql, hacemos uso de `wget` para instalar el archivo necesario para webmin desde su pagina

```
root@vbox:/tmp# wget https://www.webmin.com/download/deb/webmin-current.deb
--2026-02-01 12:53:19-- https://www.webmin.com/download/deb/webmin-current.deb
Resolving www.webmin.com (www.webmin.com)... 216.105.38.11
Connecting to www.webmin.com (www.webmin.com)|216.105.38.11|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://webmin.com/download/deb/webmin-current.deb [following]
--2026-02-01 12:53:19-- https://webmin.com/download/deb/webmin-current.deb
Resolving webmin.com (webmin.com)... 216.105.38.11
Connecting to webmin.com (webmin.com)|216.105.38.11|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://prdownloads.sourceforge.net/webadmin/webmin_2.621_all.deb [fol
lowing]
--2026-02-01 12:53:20-- https://prdownloads.sourceforge.net/webadmin/webmin_2.6
21_all.deb
Resolving prdownloads.sourceforge.net (prdownloads.sourceforge.net)... 104.18.13
.149, 104.18.12.149, 2606:4700::6812:c95, ...
Connecting to prdownloads.sourceforge.net (prdownloads.sourceforge.net)|104.18.1
3.149|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://downloads.sourceforge.net/project/webadmin/webmin/2.621/webmin
_2.621_all.deb [following]
--2026-02-01 12:53:20-- https://downloads.sourceforge.net/project/webadmin/webm
```

- Y con uso de apt-get install junto con el archivo que acabamos de descargar comenzamos la instalación de webmin

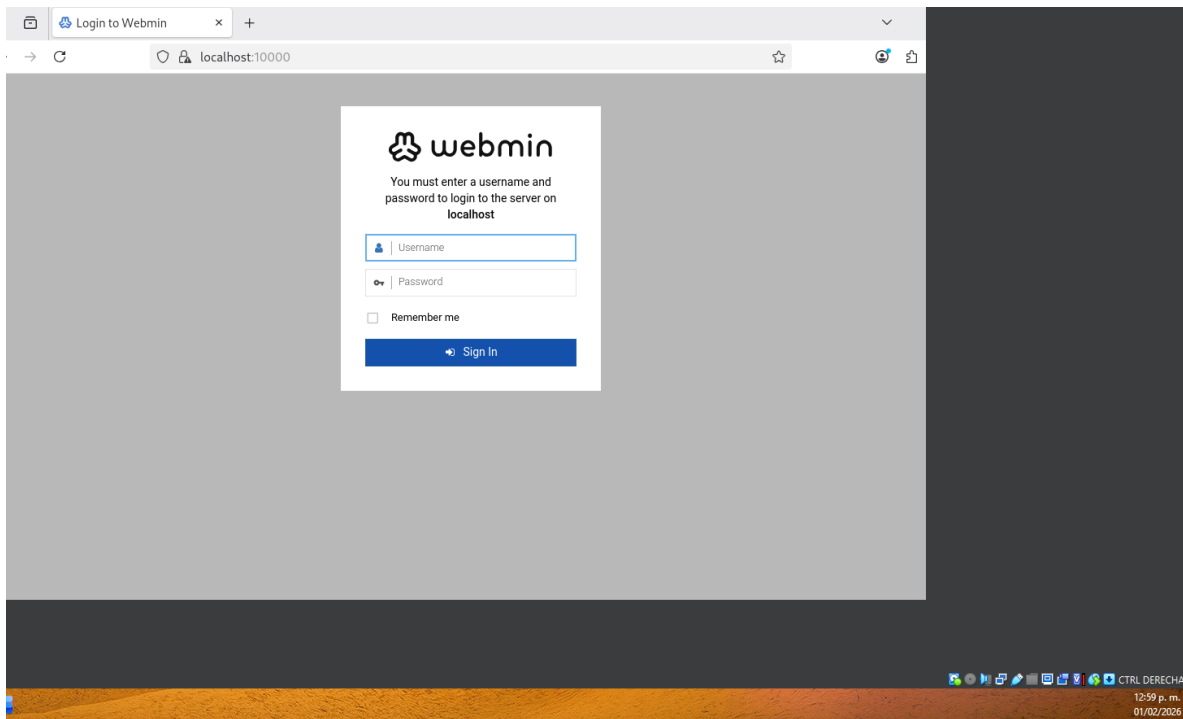
```
root@vbox:/tmp# apt-get install --install-recommends ./webmin-current.deb
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'webmin' instead of './webmin-current.deb'
The following additional packages will be installed:
  html2text libalgorithm-c3-perl libauthen-pam-perl libb-hooks-endofscope-perl
  libb-hooks-op-check-perl libclass-c3-perl libclass-c3-xs-perl
  libclass-data-inheritable-perl libclass-inspector-perl
  libclass-method-modifiers-perl libclass-singleton-perl
  libclass-xsaccessor-perl libcommon-sense-perl libdata-optlist-perl
  libdatetime-locale-perl libdatetime-perl libdatetime-timezone-perl
  libdbd-mariadb-perl libdbd-mysql-perl libdbi-perl libdevel-callchecker-perl
  libdevel-caller-perl libdevel-lexalias-perl libdevel-stacktrace-perl
  libdynamloader-functions-perl libencode-detect-perl libeval-closure-perl
  libexception-class-perl libfile-sharedir-perl libio-pty-perl libjson-xs-perl
  libmariadb3 libmodule-implementation-perl libmodule-runtime-perl
  libmro-compat-perl libnamespace-autoclean-perl libnamespace-clean-perl
  libpackage-stash-perl libpackage-stash-xs-perl libpadwalker-perl
  libparams-classify-perl libparams-util-perl
  libparams-validationcompiler-perl libreadonly-perl libref-util-perl
  libref-util-xs-perl librole-tiny-perl libsocket6-perl libspecio-perl
  libsub-exporter-perl libsub-exporter-progressive-perl libsub-identify-perl
  libsub-install-perl libsub-name-perl libsub-quote-perl
```

- Revisamos que webmin este funcionando con systemctl status

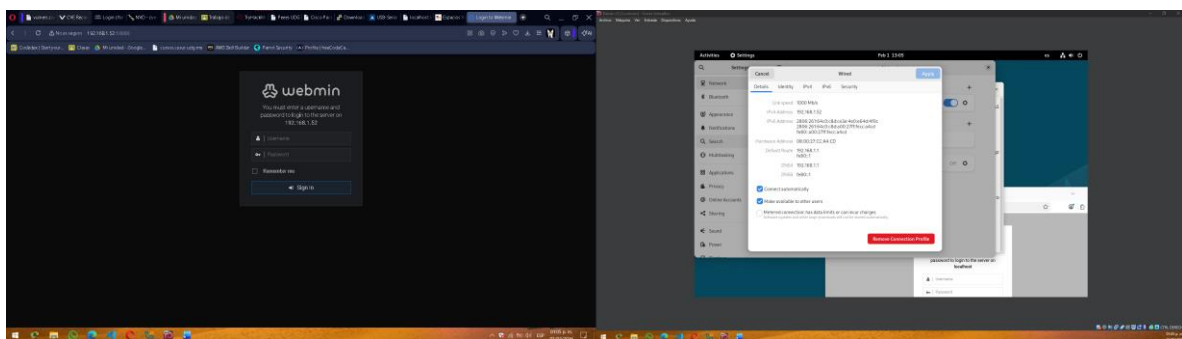
```
root@vbox:/tmp# systemctl status webmin
● webmin.service - Webmin server daemon
   Loaded: loaded (/lib/systemd/system/webmin.service; enabled; preset: enabl>
   Active: active (running) since Sun 2026-02-01 12:56:47 CST; 1min 14s ago
   Process: 6948 ExecStart=/usr/share/webmin/miniserv.pl /etc/webmin/miniserv.>
   Main PID: 6949 (miniserv.pl)
      Tasks: 2 (limit: 2277)
     Memory: 148.8M
        CPU: 6.004s
    CGroup: /system.slice/webmin.service
            └─6949 /usr/bin/perl /usr/share/webmin/miniserv.pl /etc/webmin/min>
            └─6950 /usr/bin/perl /usr/share/webmin/miniserv.pl /etc/webmin/min>

Feb 01 12:56:43 vbox systemd[1]: Starting webmin.service - Webmin server daemon>
Feb 01 12:56:44 vbox perl[6948]: pam_unix(webmin:auth): authentication failure;>
Feb 01 12:56:46 vbox webmin[6948]: Webmin starting
Feb 01 12:56:47 vbox systemd[1]: Started webmin.service - Webmin server daemon.
lines 1-16/16 (END)
```

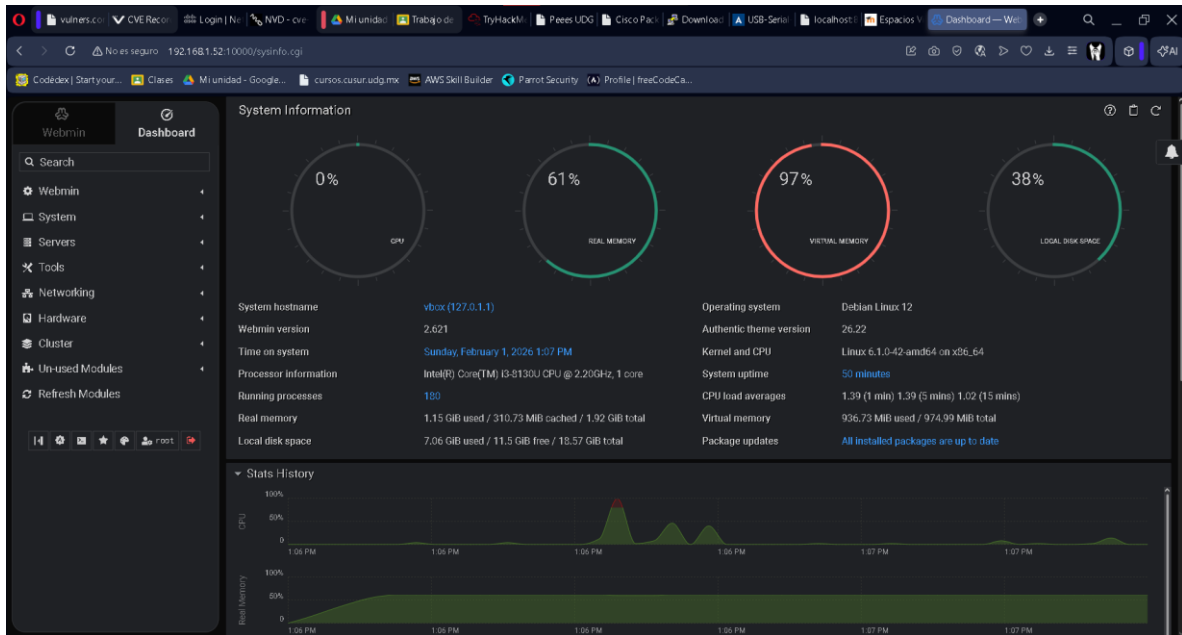
- Dentro del navegador desde la maquina ponemos en el buscador: localhost:10000 y podremos ver que webmin esta funcionando



- De la misma manera podemos ver la pagina de webmin corriendo en la maquina si usamos la ip de la maquina



- Ingresando con las credenciales root y la contraseña podemos ver el dashboard de webmin (vista desde maquina base corriend Ubuntu)



- Descargamos php con sudo apt install php

The screenshot shows a terminal window titled 'dapa@vbox:~' with the command 'sudo apt install php' being executed. The output displays the progress of installing various PHP modules. The progress bar is at 56%.

```

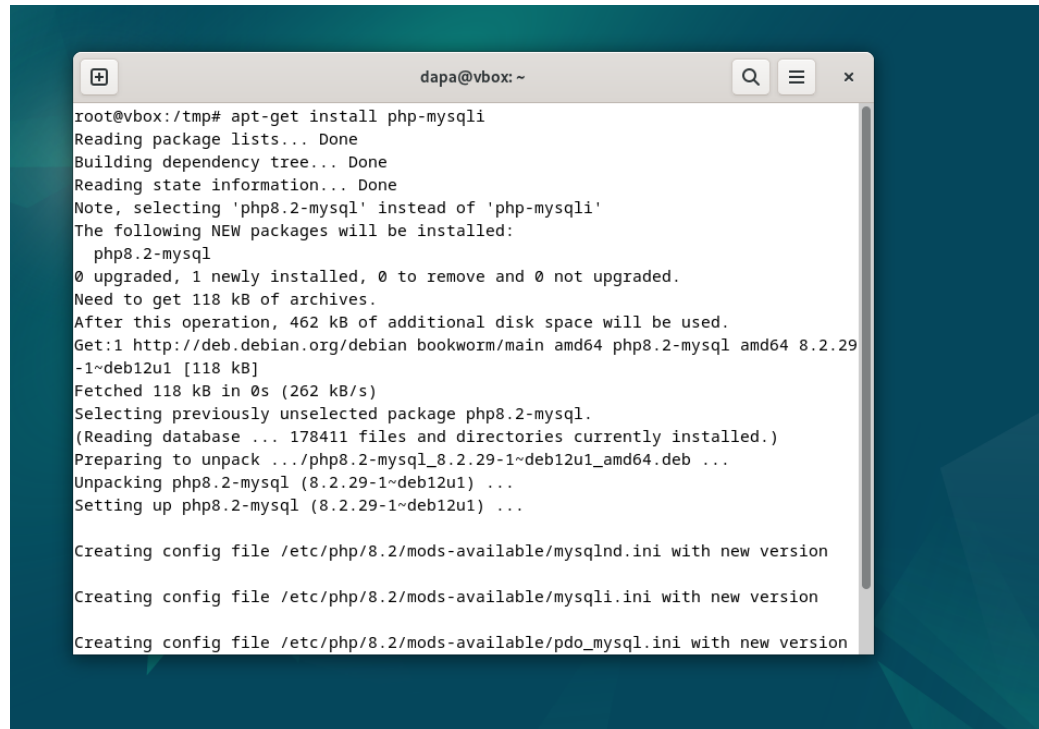
Creating config file /etc/php/8.2/mods-available/ftp.ini with new version
Creating config file /etc/php/8.2/mods-available/gettext.ini with new version
Creating config file /etc/php/8.2/mods-available/iconv.ini with new version
Creating config file /etc/php/8.2/mods-available/pdo.ini with new version
Creating config file /etc/php/8.2/mods-available/phar.ini with new version
Creating config file /etc/php/8.2/mods-available/posix.ini with new version
Creating config file /etc/php/8.2/mods-available/shmop.ini with new version
Creating config file /etc/php/8.2/mods-available/sockets.ini with new version
Creating config file /etc/php/8.2/mods-available/sysvmsg.ini with new version
Creating config file /etc/php/8.2/mods-available/sysvsem.ini with new version
Creating config file /etc/php/8.2/mods-available/sysvshm.ini with new version
Progress: [ 56% ]

```

- Una vez descargado revisamos la versión con `php -v`

```
PHP 8.2.29 (cli) (built: Jul  3 2025 16:16:05) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.2.29, Copyright (c) Zend Technologies
    with Zend OPcache v8.2.29, Copyright (c), by Zend Technologies
root@vbox:/tmp#
```

- Instalamos la librería `mysqli` con `apt-get install`

A terminal window titled 'dapa@vbox: ~' showing the command 'apt-get install php-mysqli' and its output. The output indicates that 'php8.2-mysql' will be installed instead of 'php-mysqli'. It shows the package size (118 kB) and disk space requirements (462 kB). The installation process is shown in progress, including downloading the package, unpacking it, and creating configuration files for MySQL and PDO MySQL.

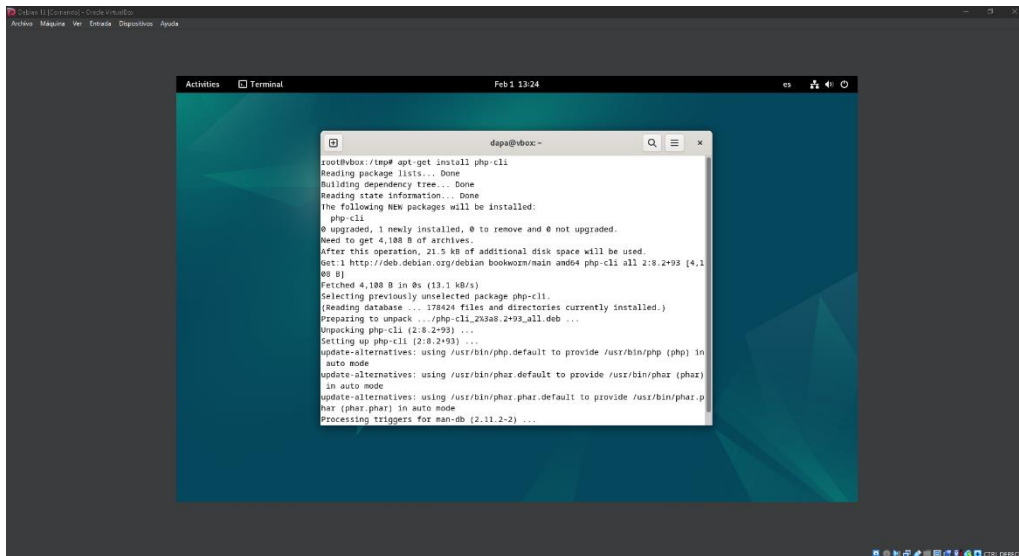
```
dapa@vbox: ~
root@vbox:/tmp# apt-get install php-mysqli
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'php8.2-mysql' instead of 'php-mysqli'
The following NEW packages will be installed:
  php8.2-mysql
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 118 kB of archives.
After this operation, 462 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian bookworm/main amd64 php8.2-mysql amd64 8.2.29-1~deb12u1 [118 kB]
Fetched 118 kB in 0s (262 kB/s)
Selecting previously unselected package php8.2-mysql.
(Reading database ... 178411 files and directories currently installed.)
Preparing to unpack .../php8.2-mysql_8.2.29-1~deb12u1_amd64.deb ...
Unpacking php8.2-mysql (8.2.29-1~deb12u1) ...
Setting up php8.2-mysql (8.2.29-1~deb12u1) ...

Creating config file /etc/php/8.2/mods-available/mysqlnd.ini with new version

Creating config file /etc/php/8.2/mods-available/mysqli.ini with new version

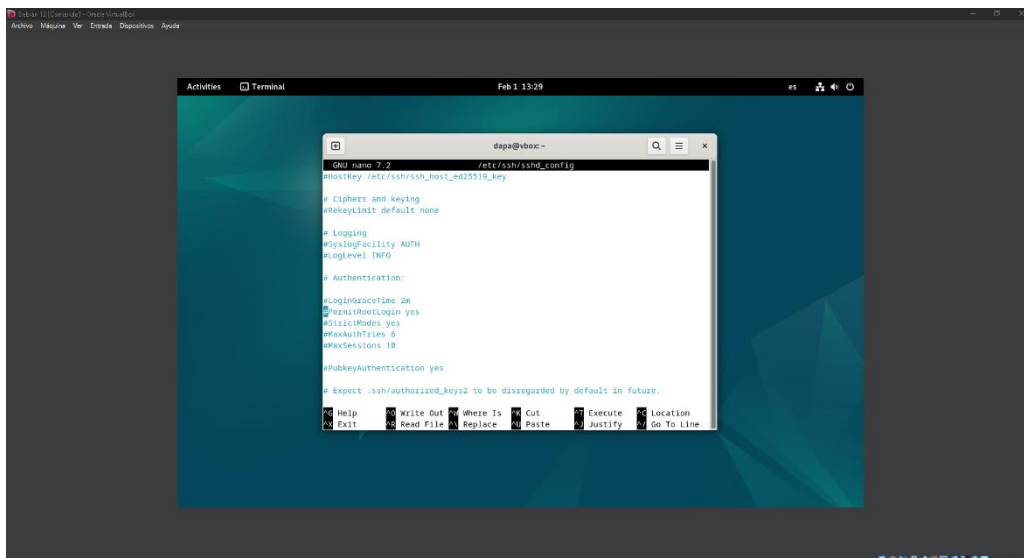
Creating config file /etc/php/8.2/mods-available/pdo_mysql.ini with new version
```

- De la misma manera con la librería php-cli



```
root@vbox:~# apt-get install php-cli
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  php-cli
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 4,188 B of archives.
After this operation, 21.5 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian bookworm/main amd64 php-cli all 2:8.2+93 [4,188 B]
Fetched 4,188 B in 0s (13.1 kB/s)
Selecting previously unselected package php-cli.
(reading database ... 178424 files and directories currently installed.)
Preparing to unpack .../php-cli_2:8.2+93_all.deb ...
Unpacking php-cli (2:8.2+93) ...
Setting up php-cli (2:8.2+93) ...
update-alternatives: using /usr/bin/php.default to provide /usr/bin/php (php) in auto mode
update-alternatives: using /usr/bin/phar.default to provide /usr/bin/phar (phar) in auto mode
update-alternatives: using /usr/bin/phar.phar.default to provide /usr/bin/phar.phar (phar.phar) in auto mode
Processing triggers for man-db (2.11.2-2) ...
```

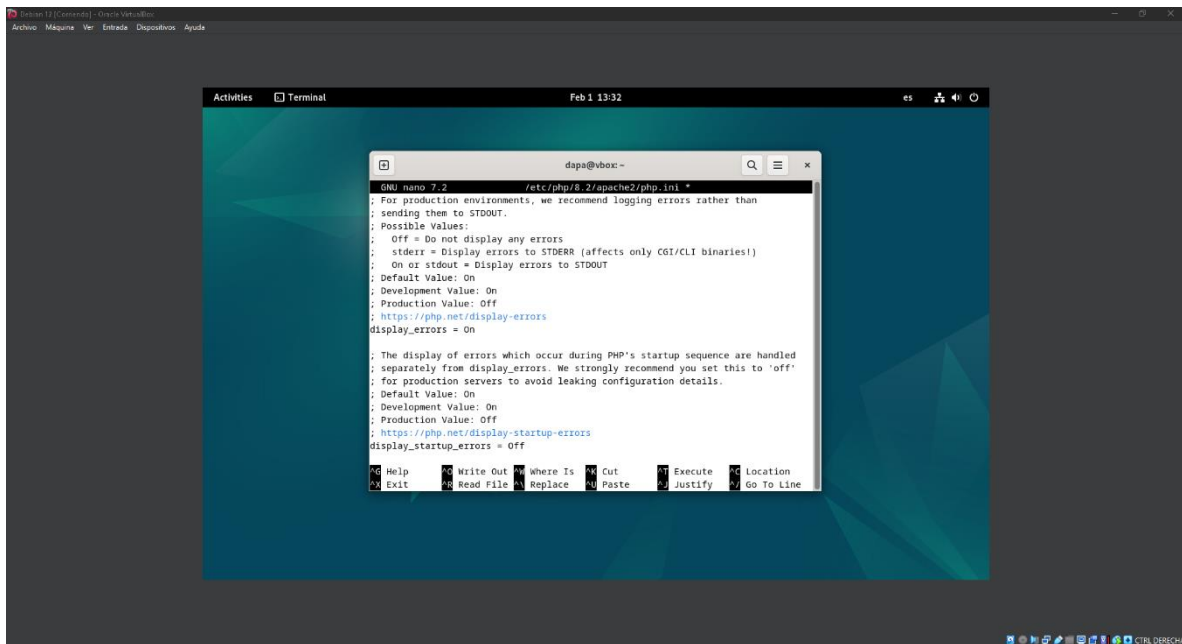
- Mediante el uso nano editamos el archivo sshd_config y editamos la línea que diga PermitRootLogin without-password por PermitRootLogin yes y reiniciamos el servicio con systemctl restart



```
GNU nano 2.9.2 /etc/ssh/sshd_config
#Port 22
#Subsystem sftp sftp-server
#X11
#X11Forwarding yes
#PermitRootLogin without-password
#PermitRootLogin yes
#MaxAuthTries 6
#MaxSessions 10
#PubkeyAuthentication yes

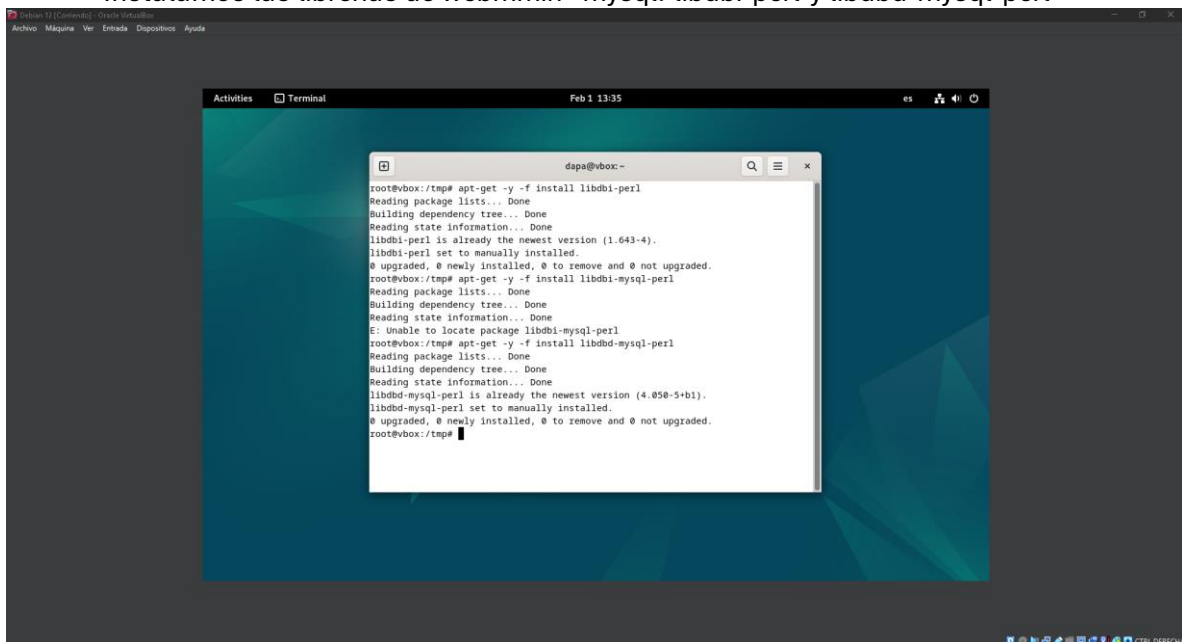
# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#Help Write Out Where To Cut Execute Location
#Exit Read File Replace Paste Justify Go To Line
```

- De nuevo con el uso de nano editamos el archivo php.ini y en la línea que contenga “display_errors” cambiamos el parámetro Off por On y reiniciamos el servicio de apache con systemctl restart



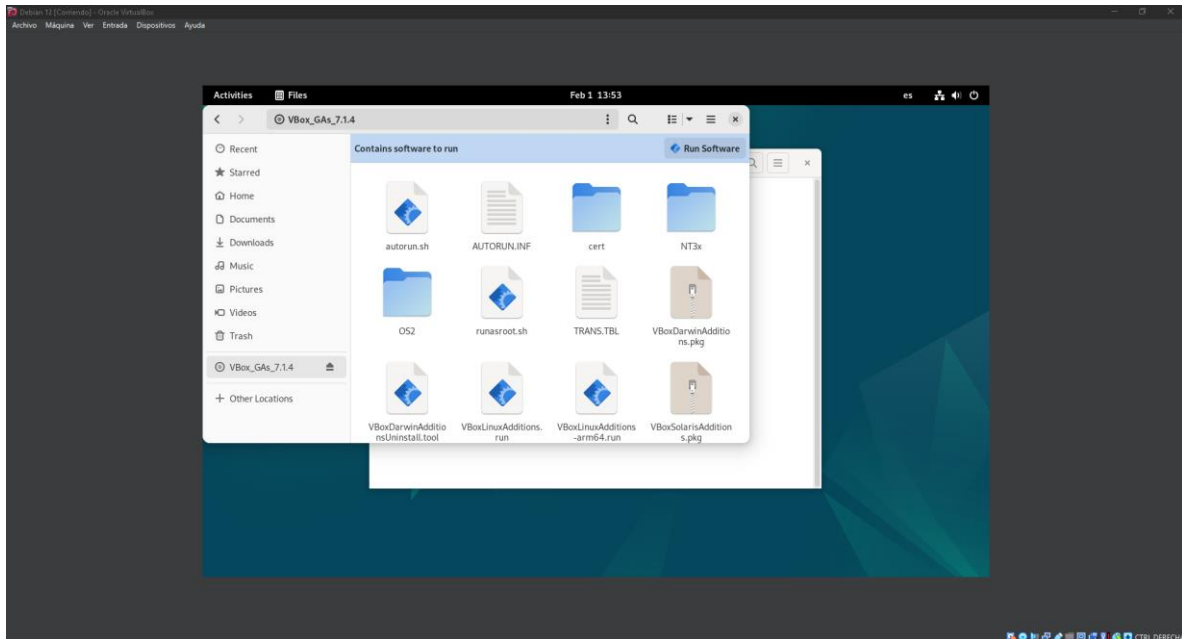
```
dapa@vbox:~  
GNU nano 2.2 /etc/php/8.2/apache2/php.ini  
; For production environments, we recommend logging errors rather than  
; sending them to STDOUT.  
; Possible Values:  
; Off = Do not display any errors  
; stderr = Display errors to STDERR (affects only CGI/CLI binaries!)  
; On or stdout = Display errors to STDOUT  
; Default Value: On  
; Development Value: On  
; Production Value: Off  
; https://php.net/display-errors  
display_errors = On  
  
; The display of errors which occur during PHP's startup sequence are handled  
; separately from display_errors. We strongly recommend you set this to 'off'  
; for production servers to avoid leaking configuration details.  
; Default Value: On  
; Development Value: On  
; Production Value: Off  
; https://php.net/display-startup-errors  
display_startup_errors = Off  
  
Help Write Out Where Is Cut Execute Location  
Exit Read File Replace Paste Justify Go To Line
```

- Instalamos las librerías de webmin -mysql: libdbi-perl y libdbd-mysql-perl

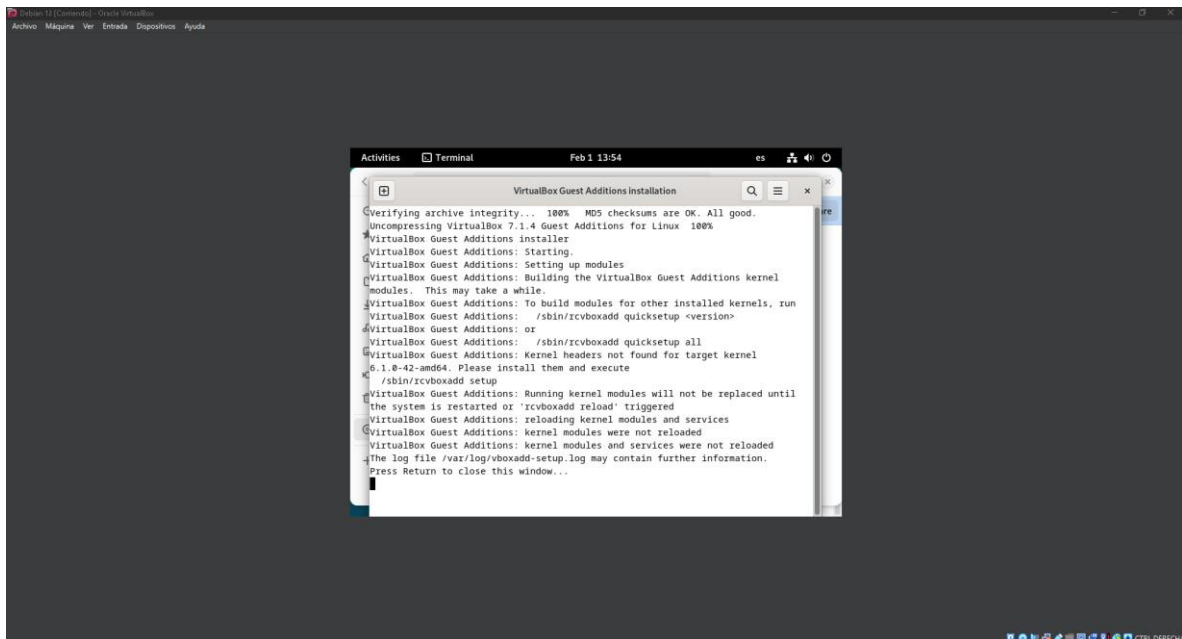


```
dapa@vbox:~  
root@vbox:/tmp# apt-get -y -f install libdbi-perl  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
libdbi-perl is already the newest version (1.643-4).  
libdbi-perl set to manually installed.  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
root@vbox:/tmp# apt-get -y -f install libdbd-mysql-perl  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
E: Unable to locate package libdbi-mysql-perl  
root@vbox:/tmp# apt-get -y -f install libdbd-mysql-perl  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
libdbd-mysql-perl is already the newest version (4.050-5+b1).  
libdbd-mysql-perl set to manually installed.  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
root@vbox:/tmp#
```

- Desde el menú de VirtualBox damos click en Dispositivos>Insertar imagen de CD de los complementos del invitado, seguido de eso, en Ubuntu, entramos al explorador de archivos y damos clic en el CD de Vbox y damos clic en Run Software.



- Solo queda esperar a que termine



- Al final desde el menú de VirtualBox editamos los parámetros de portapapeles compartido y Arrastrar y soltar ambos en Bidireccional